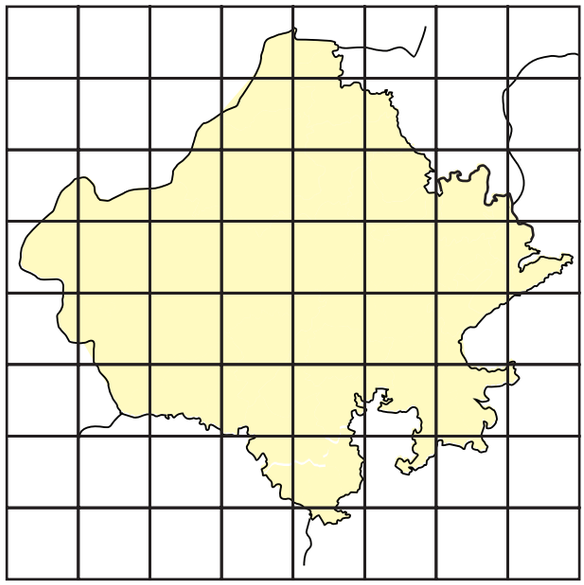
equal to the square of the ratios between the scales of the original and enlarged or reduced maps.

Graphical Method

Graphically maps can be enlarged or reduced with the help of similar squares.

The square method is the most common and simplest method for enlargement and reduction of maps. In order to enlarge a map, cover the original map with a set of squares of equal sides. The side of the squares has to be enlarged proportionally to that the original map. The side of the square of the new map has to be determined using the formula.

Step 1: Draw a network of squares on the original map, each side being 1 cm. in length.



Rajasthan R.F=1:16,000,000

Scale of the new map $= \frac{\text{New scale}}{\text{old scale}} \times$ Side of the square of the original map.

Example: 2

This is a map of Rajasthan drawn on a scale of 1/16,000,000 and is to be enlarged on the scale of 1/8,000,000.



To enlarge the given map on the scale of 1/8,000,000:

Step 2: Calculation

When the scale is 1/16,000,000 the side of the small square is one cm.

For Scale of the map 1/16,000,000 side of the small square =1 cm

Therefore, If scale is to be 1/8,000,000 side of the small square of new map = x

$$X = \frac{1/8,000,000}{1/16,000,000} \text{ x } 1 \text{ cm}$$

$$X = \frac{1 \text{ x } 16,000,000}{1 \text{ x } 8,000,000} = 2\text{cm}$$

When the scale is 1/8,000,000 the side of the small square will be 2cmNow draw a network of squares, each side measuring 2 cm. The number of squares will be the same as on the original map. Now transfer the outline of original map on the enlarged map square by square.

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